

SKILL WEEK 9

Task 1:

Code:

```
#include <stdio.h>
#include <unistd.h>
#include <fcntl.h>

int main() {
    int fd = open("input.txt", O_RDONLY);
    if (fd < 0) {
        perror("input.txt");
        return 1;
    }

    dup2(fd, STDIN_FILENO);
    close(fd);

    char buf[100];
    ssize_t n = read(STDIN_FILENO, buf, sizeof(buf) - 1);
    if (n < 0) {
        perror("read");
        return 1;
    }

    buf[n] = '\0';
    printf("Input from redirected file:\n%s", buf);
    return 0;
}
```

OUTPUT:

```
ashishreddy@Ashishs-MacBook-Air ~ % nano 9.1.c
ashishreddy@Ashishs-MacBook-Air ~ % echo "Hello from input file" > input.txt
ashishreddy@Ashishs-MacBook-Air ~ % gcc 9.1.c -o 9.1
ashishreddy@Ashishs-MacBook-Air ~ % ./9.1
Input from redirected file:
Hello from input file
```

SKILL WEEK 9

Task 2:

Code:

```
#include <stdio.h>
#include <unistd.h>
#include <fcntl.h>

int main() {
    int fd = open("output.txt", O_WRONLY | O_CREAT | O_TRUNC, 0644);
    if (fd < 0) {
        perror("output.txt");
        return 1;
    }

    dup2(fd, STDOUT_FILENO);
    close(fd);

    printf("This output is redirected to a file.\n");
    printf("Output redirection completed.\n");

    return 0;
}
```

OUTPUT:

```
ashishreddy@Ashishs-MacBook-Air ~ % nano 9.2.c
ashishreddy@Ashishs-MacBook-Air ~ % gcc 9.2.c -o 9.2
ashishreddy@Ashishs-MacBook-Air ~ % ./9.2
ashishreddy@Ashishs-MacBook-Air ~ % cat output.txt
This output is redirected to a file.
Output redirection completed.
```

SKILL WEEK 9

Task 3:

Code:

```
#include <stdio.h>
#include <unistd.h>
#include <fcntl.h>

int main() {
    int fd = open("append.txt", O_WRONLY | O_CREAT | O_APPEND, 0644);
    if (fd < 0) {
        perror("append.txt");
        return 1;
    }

    dup2(fd, STDOUT_FILENO);
    close(fd);

    printf("New line appended to the file.\n");
    return 0;
}
```

OUTPUT:

```
ashishreddy@Ashishs-MacBook-Air ~ % nano 9.3.c
ashishreddy@Ashishs-MacBook-Air ~ % gcc 9.3.c -o 9.3
ashishreddy@Ashishs-MacBook-Air ~ % ./9.3
ashishreddy@Ashishs-MacBook-Air ~ % ./9.3
ashishreddy@Ashishs-MacBook-Air ~ % cat append.txt
New line appended to the file.
New line appended to the file.
```

SKILL WEEK 9

Task 4:

Code:

```
#include <stdio.h>
#include <unistd.h>
#include <fcntl.h>

int main() {
    int fd = open("error.txt", O_WRONLY | O_CREAT | O_TRUNC, 0644);
    if (fd < 0)
        return 1;

    dup2(fd, STDERR_FILENO);
    close(fd);

    fprintf(stderr, "Error: file could not be opened.\n");
    fprintf(stderr, "stderr redirection completed.\n");

    return 0;
}
```

OUTPUT:

```
ashishreddy@Ashishs-MacBook-Air ~ % nano 9.4.c
ashishreddy@Ashishs-MacBook-Air ~ % gcc 9.4.c -o 9.4
ashishreddy@Ashishs-MacBook-Air ~ % ./9.4
ashishreddy@Ashishs-MacBook-Air ~ % cat error.txt
Error: file could not be opened.
stderr redirection completed.
```